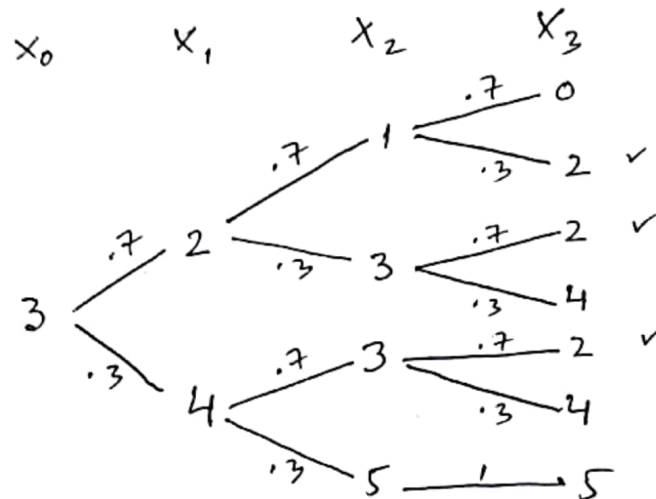


①

1.

$$P(X_3 = 2) = .7 \times .7 \times .3 + .7 \times .3 \times .7 + .3 \times .7 \times .7$$

$$= 0.441$$

2.

	0	1	
0	2	14	16
1	14	15	29

$$P = \begin{matrix} & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 1/8 & 7/8 \\ 14/29 & 15/29 \end{pmatrix} \end{matrix}$$

3.

$$P^2 = \begin{pmatrix} .2 & .1 & .7 \\ .3 & .3 & .4 \\ .2 & .3 & .5 \end{pmatrix} \begin{pmatrix} .2 & .1 & .7 \\ .3 & .3 & .4 \\ .2 & .3 & .5 \end{pmatrix} = \begin{pmatrix} - & - & - \\ - & - & - \\ - & 0.26 & - \end{pmatrix}$$

$$P(X_2 = 2 | X_0 = 3) = P_{32}^{(2)} = 0.26$$

$$\underline{\underline{4.}} \quad P = \begin{matrix} & \begin{matrix} I & II \end{matrix} \\ \begin{matrix} I \\ II \end{matrix} & \begin{pmatrix} 0.70 & .30 \\ .20 & .80 \end{pmatrix} \end{matrix}$$

$$\pi P = \pi$$

$$\Rightarrow (\pi_1, \pi_2) \begin{pmatrix} .70 & .30 \\ .20 & .80 \end{pmatrix} = (\pi_1, \pi_2)$$

$$\therefore \begin{cases} .7\pi_1 + .2\pi_2 = \pi_1 \dots (A) \\ .3\pi_1 + .8\pi_2 = \pi_2 \dots (B) \end{cases} \quad \left| \begin{array}{l} \pi_1 + \pi_2 = 1 \\ \therefore \pi_2 = 1 - \pi_1 \dots (C) \end{array} \right.$$

(C) in (A)

$$.7\pi_1 + .2 - .2\pi_1 = \pi_1$$

$$\therefore \pi_1 = 0.4$$

$$\therefore \pi_2 = 1 - 0.4 = 0.6$$

Comment: In the long run, 40% of customers will buy brand I and 60% will buy II.

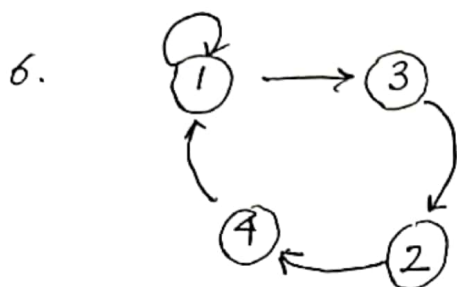
5. Marginal distribution of X_1 :

$$\underline{a}^{(1)} = \underline{a}^{(0)} P = (0.13, 0.30, 0.57)$$

Marginal of X_2 :

$$\underline{a}^{(2)} = \underline{a}^{(0)} P^2 = \underline{a}^{(1)} P = (0.130, 0.317, 0.553)$$

(3)



irreducible (and finite)
 So, all states are
 positive recurrent.

(other arrows are
 not needed for
 decision)

$P_{11}^{(1)} = 0.4$ (diagonal
 element positive)

So, period of 1 is 1.

$\therefore$ other periods are 1.
 (all communicating).

7. Equivalence classes:

$\{2, 4, 5\}$ (matrix within matrix)

$\{3, 6\}$ open, so transient, period
 irrelevant

$\{1\}$ open, $\therefore$ transient, period
 irrelevant

$\{2, 4, 5\}$ is closed and finite. So,
 all these states 2, 4 and 5 are positive
 recurrent. $P_{22}^{(1)} = 0.3$ (diagonal element +ve).
 So, period of 2 is 1. So, period of
 4 and 5 is 1.

— X —